

RESEARCH WHITE PAPER

Chess Engine Neutralisation

Prophylaxis, Decision-Space Compression, and Strategic Control
Against Computational Opponents

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Version 1.0 — 2 October 2026

Status. This is a conceptual research white paper, not a peer-reviewed empirical study. It consolidates ideas developed through repeated human–AI analysis of chess positions and engine play, relates them to established positional chess and computer-chess literature, and proposes falsifiable experiments. Claims of novelty are deliberately scoped: established chess principles are identified as such, while the paper’s distinctive contribution is presented as a synthesis and research programme rather than as a proven priority claim. **Suggested**

citation. Venkataraman, A. and GPT-5.6 Luna. “Chess Engine Neutralisation: Prophylaxis, Decision-Space Compression, and Strategic Control Against Computational Opponents.” Research White Paper, Version 1.0, October 2026.

Abstract

Chess has traditionally framed prophylaxis as the prevention of an opponent’s plan, while modern computer chess has produced opponents whose tactical and strategic search is beyond ordinary human calculation. This paper proposes **Chess Engine Neutralisation**: a practical and conceptual framework in which prophylactic play is extended from move-level threat prevention to deliberate manipulation of an opponent’s **effective decision space**. The central objective is not to calculate more deeply than a superhuman engine, but to create positions in which fewer of the engine’s available actions remain strategically useful.

The paper introduces five core constructs: **effective decision space**, **decision-space compression**, **critical resources**, **control advantage**, and the **neutralisation–conversion separation**. It integrates these constructs into an operational six-stage protocol: **Neutralise – Stabilise – Probe – Induce – Strike – Convert**. It also proposes opponent-model dependence: neutralisation against a human, a human-like model such as Maia, and a superhuman engine such as Stockfish should target different structures of choice.

The framework is intentionally conservative about novelty. Prophylaxis, restraint, piece restriction, squeezing, and worst-piece improvement are established ideas with deep roots in positional chess. The proposed contribution is the synthesis of these ideas around a computational-opponent objective and their translation into testable measures of decision-space compression. A 2026 manuscript independently demonstrates that decision-space entropy can be estimated in large chess corpora and is associated with human error, creating a useful methodological bridge for testing the framework. The paper therefore closes with a rigorous empirical agenda involving multi-PV engine analysis, entropy and candidate-set metrics, critical-resource tagging, Maia/Stockfish comparison, and controlled ablation studies.

Keywords: chess engines, prophylaxis, positional chess, Stockfish, Maia, decision space, entropy, restraint, strategic control, human-AI interaction, computational chess

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1 Introduction

The arrival of superhuman chess engines changed the nature of practical chess analysis. Human players no longer face a purely human adversary when training with or playing against systems such as Stockfish. The question is not simply how to find a good move, but how a human decision-maker should think when the opposing search process is vastly deeper, faster, and more tactically precise than human calculation.

The obvious response is to imitate the machine: calculate further, memorize more opening theory, or search for tactical complications. That response has limits. A human cannot meaningfully reproduce the computational breadth of a modern engine. Current Stockfish releases are explicitly described by the project as stronger than any human, with Stockfish 18 reporting substantial gains over its predecessor and continuing to dominate engine competition [7]. AlphaZero demonstrated a different route to superhuman chess: self-play reinforcement learning combined with search can discover strong chess without handcrafted human heuristics [4].

This paper asks a different question:

If a human cannot out-calculate a computational opponent, can the human instead shape the position so that the opponent has fewer strategically useful decisions to exploit?

The proposed answer is the framework of **Chess Engine Neutralisation**.

The key move in the framework is conceptual. Classical prophylaxis asks, “What does my opponent want to do?” Nimzowitsch placed prevention at the heart of positional play, including the prevention of freeing moves and harmful positional options [1]. Chess pedagogy continues to teach the same habit: identify the opponent’s plan and remove it before pursuing one’s own [8].

Engine neutralisation generalises this question:

What does this opponent need to make possible, and which of those enabling conditions can I remove before they become operational?

The result is not passive defence. It is **strategic control**: an attempt to make the opponent’s strongest machinery progressively less relevant by constraining its useful action space while preserving the integrity of one’s own position.

2 Scope and Epistemic Status

This paper makes three different kinds of statement, which must not be conflated.

1. **Established chess knowledge.** Prophylaxis, restraint, restriction, overprotection, improving the worst piece, squeezing small advantages, and denying counterplay have substantial historical and pedagogical precedents.
2. **Distinctive synthesis.** The paper unifies these ideas around the explicit objective of reducing a computational opponent’s strategically useful decision space. The terms “engine neutralisation,” “decision-space compression” in this strategic sense, “critical-resource targeting,” and “neutralisation–conversion separation” are used here as an integrated framework.
3. **Research hypotheses.** The paper proposes quantitative metrics and empirical claims that are not established by the present manuscript. These require testing across large game samples and controlled experiments.

The paper should therefore be read as a *research framework and white paper*, not as proof that the proposed mechanism improves human results against engines.

3 Background

3.1 Prophylaxis and restraint

Prophylaxis entered positional chess vocabulary through Aron Nimzowitsch’s *My System*. The concept is commonly described as taking precautionary measures to neutralise strategically harmful options; in Nimzowitsch’s treatment, outer prophylaxis focuses on the opponent’s plans and freeing moves, while inner prophylaxis protects one’s own important points and coordination [1].

This matters because a prophylactic move often has *negative value* for the opponent rather than obvious immediate value for oneself. A quiet move may not attack a piece, advance a pawn, or increase an engine’s static evaluation dramatically. Its value lies in removing a future branch of play.

That observation provides the historical foundation for the present framework, but it does not by itself constitute novelty. The novelty question begins when the prevented branches are treated as elements of a measurable decision space.

3.2 Computer chess as a different opponent class

Classical computer chess grew from minimax and search formulations, through systems such as Deep Blue, toward today’s neural and hybrid engines [2, 3]. AlphaZero demonstrated that superhuman chess need not rely on the traditional architecture of handcrafted evaluation and massive brute-force search [4].

Modern Stockfish is an especially relevant computational opponent because it combines highly optimized search with a learned NNUE evaluation architecture. Stockfish 18, released in January 2026, reports up to a 46 Elo-point gain versus Stockfish 17 in its testing and describes performance beyond human level across hardware classes [7].

For the present framework, however, the internal engine architecture is secondary. The crucial fact is that the opponent can evaluate and search far more candidate continuations than a human can. The human problem is therefore one of **resource asymmetry**.

3.3 Human-like engines and Maia

Not all chess AIs model the same opponent. Maia was designed specifically to predict human moves rather than merely select near-optimal moves, using human game data and tunable playing levels [5]. Later work extended the approach to modelling individual human behaviour [6].

This motivates an important principle of the present framework:

Neutralisation is opponent-model dependent. The resource that must be neutralised against a human need not be the resource that matters against Maia, and neither is identical to the resource that matters against Stockfish.

Against a human, intention and practical psychology may dominate. Against Maia, behavioural regularities and human-likely choices become relevant. Against Stockfish, the target is more abstract: the set of strategically useful continuations generated by a superhuman search process.

4 From Prophylaxis to Engine Neutralisation

4.1 The central shift

Traditional prophylaxis can be represented as:

$$\text{Identify opponent plan} \rightarrow \text{Prevent plan} \rightarrow \text{Continue own plan.} \quad (1)$$

Engine neutralisation extends this into:

$$\begin{aligned} &\text{Model opponent resources} \rightarrow \text{Restrict enabling conditions} \rightarrow \text{Reduce useful action space} \\ &\rightarrow \text{Stabilise} \rightarrow \text{Improve} \rightarrow \text{Convert.} \end{aligned} \quad (2)$$

The difference is subtle but important. The target is no longer only a named plan such as a pawn break or piece manoeuvre. The target is the **resource structure** that gives the opponent many strong options.

4.2 Why “neutralisation” rather than “beating”

The framework deliberately distinguishes between **neutralising an opponent’s advantage-producing mechanisms** and **winning the game**. A human may be capable of reaching and repeatedly maintaining a strategically resilient position against an engine without being able to convert that resilience into a win.

This distinction emerged repeatedly in practical analysis: stable positions can sometimes be held, including by repeated moves or repetition, while attempts to “play on” can expose the gap between neutralisation skill and technical conversion skill. The observation is anecdotal, not statistical, but it motivates a formal research question:

Can resistance to a stronger engine and conversion of an advantage against that engine be measured as separable abilities?

5 Core Constructs

5.1 Effective decision space

Let a chess position be S and let $\mathcal{C}(S)$ be a set of legal candidate moves considered relevant under an engine analysis policy. Not every legal move is strategically meaningful. We therefore define an **effective decision space** as the set of actions whose estimated utility lies within a chosen tolerance of the best available action, or more generally a probability-weighted distribution over candidate actions.

A simple threshold formulation is:

$$\text{EDS}_\tau(S) = \{m \in \mathcal{C}(S) : \mathcal{E}(m \mid S) \geq \mathcal{E}^*(S) - \tau\}, \quad (3)$$

where $\mathcal{E}^*(S)$ is the best engine evaluation for the side to move and τ is a tolerance in centipawns or another normalised unit.

This is deliberately a *research definition*, not a claim that one threshold is universally correct.

5.2 Decision-space compression

Suppose an opponent moves from position S to S' after our move m . A simple compression ratio is:

$$\text{DSC}_\tau(m; S) = 1 - \frac{|\text{EDS}_\tau(S')|}{|\text{EDS}_\tau(S)|}. \quad (4)$$

Positive values indicate fewer near-optimal candidates after the intervention. Negative values indicate that the opponent’s effective candidate set has expanded.

However, candidate count is crude. A position with five candidates of nearly equal value may be more complex than a position with five candidates where one is clearly superior. A probability-weighted measure is therefore preferable.

One natural construction, inspired by Luce/McFadden-style choice models, is:

$$p_i = \frac{\exp(E_i/T)}{\sum_j \exp(E_j/T)}, \quad H(S) = - \sum_i p_i \log p_i, \quad (5)$$

where E_i are candidate evaluations and T is a temperature parameter. A 2026 empirical manuscript has independently used this general family of entropy measures on 9,114 online games, reporting an association between decision-space entropy and human error [9]. Because that work is recent and not yet peer-reviewed, it should be regarded here as related methodology rather than settled evidence.

Under this formulation, strategic compression can be represented by:

$$\Delta H_{opp}(m; S) = H_{opp}(S) - H_{opp}(S'). \quad (6)$$

A positive value means that our move makes the opponent’s useful choices more concentrated.

5.3 Critical resource

The framework’s strongest practical heuristic is that the most important enemy target is often not the most valuable piece. It is the **critical resource**: the piece, square, pawn break, file, exchange pattern, or coordination link whose preservation is necessary for a strategic plan to function.

Formally, for a candidate strategic plan P , define a critical resource r as a feature whose removal causes a disproportionately large reduction in the plan’s feasible continuation set:

$$\text{CR}(r; P) = \frac{\text{feasible continuations of } P}{\text{feasible continuations of } P \text{ after removing } r}. \quad (7)$$

Again, this is a proposed research quantity rather than a standard chess metric.

The practical question becomes:

Which enemy piece or square must remain functional for the opponent’s plan to work?

This is stronger than merely asking which enemy piece is active.

5.4 Control advantage versus evaluation advantage

Engine evaluation compresses a rich position into a scalar or outcome probability. Practical controllability is different. Consider two positions with identical engine evaluation +0.5.

Position A may contain numerous tactical resources for both sides, dynamic imbalances, and many near-equal engine choices. Position B may contain very few viable counterplay mechanisms and a highly stable structure in which the stronger side can improve incrementally.

The framework therefore distinguishes:

$$\text{Evaluation advantage} \neq \text{Control advantage.} \quad (8)$$

A useful future composite index could combine evaluation, own-position stability, and opponent decision-space compression:

$$C(S) = \alpha \mathcal{E}(S) + \beta R_{self}(S) + \gamma \Delta H_{opp}(S), \quad (9)$$

where R_{self} is a proposed resilience metric and α, β, γ require empirical calibration.

The point is not to replace engine evaluation, but to augment it with a measure of **position control**.

5.5 Neutralisation versus conversion

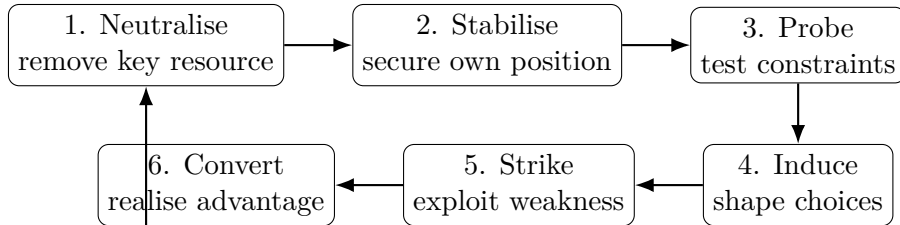
The framework decomposes practical strength against a superior engine into two stages:

1. **Neutralisation:** prevent the opponent from obtaining stable, high-value sources of counter-play or tactical leverage.
2. **Conversion:** once the position is controlled, increase the advantage and realise it technically.

This separation is important because failure to convert does not imply failure to neutralise. Conversely, a player may create a winning engine evaluation by taking risks that would not be classed as good neutralisation.

6 The Neutralisation Protocol

The framework’s practical cycle is:



6.1 1. Neutralise

Ask what the opponent would play if given a free move. Identify the tactical motif, freeing pawn break, activation route, invasion square, exchange pattern, or critical piece that enables that plan.

The preferred prophylactic move is one that removes the resource with minimal damage to one’s own structure or activity.

6.2 2. Stabilise

Before seeking more advantage, remove tactical defects in your own position. A neutralised opponent can still exploit a self-created weakness. Stabilisation therefore protects against the common failure mode of confusing passivity with control.

6.3 3. Probe

Once the main resource is restricted, make a low-risk improvement that tests what counterplay remains. Examples include improving the worst piece, controlling a critical square, or preparing an exchange that preserves structural superiority.

6.4 4. Induce

The player may now create a position in which the opponent’s remaining choices are increasingly constrained. The objective is not to force an artificial weakness, but to make the opponent commit in a position where the original freedom has already been reduced.

6.5 5. Strike

The tactical or positional strike comes after constraint, not before it. When the critical resource has been removed and the defender’s pieces are tied to passive duties, a previously modest weakness can become exploitable.

6.6 6. Convert

Simplify where appropriate, transition to a technically favourable endgame, or maintain the squeeze until a concrete tactical opportunity appears. Conversion is a distinct phase, not merely the final move of neutralisation.

7 Decision-Space Engineering

The most general interpretation of the framework is to treat chess as an interaction between **position state** and **decision space**.

Let S_t be the state after move t . A move is then not only a transition in material, geometry, and evaluation; it is also a transformation of the opponent’s next action set:

$$S_t \xrightarrow{m_t} S_{t+1}, \quad \text{EDS}_{\text{opp}}(S_t) \rightarrow \text{EDS}_{\text{opp}}(S_{t+1}). \quad (10)$$

This leads to the central thesis:

A strong prophylactic move can be understood as a state transition that preserves or improves the player’s own viability while shrinking the opponent’s future set of strategically useful actions.

This is why the phrase **decision-space engineering** is useful. It does not imply that chess is literally an engineering problem. It provides a model for analysing how a player can manipulate the structure of choices rather than merely respond to individual moves.

7.1 A control-loop interpretation

The framework can also be read as a control loop:

$$\text{Observe} \rightarrow \text{Model intent} \rightarrow \text{Constrain} \rightarrow \text{Observe new state} \rightarrow \text{Update model}. \quad (11)$$

The important feedback signal is not only engine evaluation. It is also how the opponent’s viable options have changed.

This suggests a new practical training habit:

After every prophylactic move, ask not only “Did the evaluation improve?” but also “What became impossible, impractical, or unattractive for the opponent?”

8 The Critical-Resource Heuristic

A recurrent positional discovery in our analysis was that the enemy’s strategic system often depends on one surprisingly modest component.

A queen may retreat safely, yet in doing so abandon coordination. A bishop may remain materially intact but lose its only useful diagonal. A knight may have many legal squares but lose the one square that makes the attacking structure coherent. A pawn break may be legal but become strategically harmless because the supporting piece has been removed.

This motivates a five-question heuristic:

1. What is the opponent’s intended plan?
2. Which piece, square, break, or exchange makes that plan operational?
3. Can I remove or constrain that resource without creating a worse weakness?
4. After I do so, which of my own pieces becomes more useful?
5. Can I improve while preserving the restriction?

The emphasis is on **functional necessity**, not piece value.

8.1 Illustrative motif from practical analysis

A representative position considered in our earlier analysis was:

2r2rk1/pp1q1ppp/2np4/8/2P5/2N1PN2/PP3PPP/2RQ1RK1 w - - 0 1

The analysis centred on the relationship between Black’s queen placement and the functional mobility of White’s f3-knight. The important question was not whether the knight was attacked immediately, but whether Black could choose a queen square that made the knight’s intended advance significantly less effective. The example is useful precisely because it illustrates the critical-resource concept: a piece can be strategically constrained through the opponent’s coordination rather than through direct material attack.

This position is presented as an *illustrative analytical motif*, not as a formally verified engine theorem. A rigorous paper should reproduce such claims with fixed engine versions, depths, multiple principal variations, and independent analysis.

9 The +0.5 Squeeze and Positional Control

A recurring practical philosophy underlying the framework is the value of a small, stable advantage. Call this the **+0.5 squeeze** as a shorthand for positions in which the stronger side has a modest static advantage but high structural control.

The key insight is that a small evaluation advantage can have very different practical meanings depending on the position’s volatility and counterplay.

This does not imply that a controlled +0.5 is objectively “better” than a volatile +0.5 when the engine evaluates both equally. It means that a second dimension of analysis may be useful for human decision-making: **how much freedom does the opponent retain?**

The framework therefore treats squeezing as a control process rather than simply an accumulation of static advantages:

Table 1: Two positions with the same nominal engine evaluation can have different control profiles.

	High-volatility +0.5	Controlled +0.5
Opponent choices	Many near-equal tactical resources	Few strategically useful resources
Own risk	Significant counterplay exposure	Low counterplay exposure
Piece coordination	Dynamic and unstable	Stable and improving
Conversion path	Unclear	Gradual and repeatable
Framework interpretation	Evaluation advantage	Evaluation + control advantage

$$+0.5 \rightarrow \text{restrict} \rightarrow \text{improve} \rightarrow \text{restrict} \rightarrow \text{improve} \rightarrow \text{convert}. \quad (12)$$

10 Opponent-Model Dependence

The same board position can require different neutralisation logic depending on who or what is on the other side.

Table 2: Opponent-model view of neutralisation.

Opponent	Primary modelling target	Neutralisation emphasis
Human	Intent, plans, tactical vision, psychology, time management	Remove the plan, simplify choice, exploit predictable weaknesses
Maia	Human-like move distribution at a chosen skill level	Constrain plausible human continuations and exploit modelled tendencies
Stockfish-class engine	Computationally generated tactical and strategic resources	Reduce the number and diversity of high-quality continuations; preserve structural resilience

Maia’s research programme is especially relevant because it shows that modelling human move choice is a distinct task from selecting optimal moves [5, 6]. The present framework adopts that distinction and applies it to prophylactic practice.

11 Empirical Observations and Practical Motifs

The framework emerged not from a single experiment, but from repeated position-by-position analysis. Several recurring observations deserve preservation as hypotheses.

11.1 Neutralisation can succeed before conversion

A player may repeatedly reach a position where the engine is unable to make meaningful progress without the human player making a concession. If the human instead forces a repetition, the position can remain stable. If the human deliberately avoids repetition, the engine may eventually exploit endgame precision or a latent weakness.

The practical lesson is that “I can hold this” and “I can win this” are different propositions. The research analogue is a two-axis capability model: **resilience** and **conversion**.

11.2 Restriction can matter more than attack

A strategically important enemy piece may remain on the board and even avoid direct attack, yet become ineffective because the squares, pawn breaks, or supporting pieces it needs are unavailable.

This is a more general form of restraint: attack the *function* rather than the *piece*.

11.3 Safe queen moves can have strategic cost

A queen can be materially safe but strategically displaced. A retreat to a seemingly secure square can disconnect the queen from a supporting piece, break a coordination motif, or make another enemy piece passive.

This observation suggests that future metrics should track not only raw legal mobility but **functional mobility**: the set of moves that preserve strategic coordination.

11.4 Engine strength can be made less relevant

The framework does not posit that Stockfish becomes weaker in a constrained position. Rather, its claim is that the value of its computational advantage may diminish when there are fewer strategically useful branches to exploit.

That distinction is essential. Neutralisation is not an attack on the engine’s calculation; it is an attempt to alter the game tree presented to that calculation.

12 Formal Research Hypotheses

The framework yields several falsifiable hypotheses.

12.1 H1: Resource-space hypothesis

Controlling for engine evaluation, positions with smaller effective opponent decision spaces should be more resilient to human play than positions with larger decision spaces.

12.2 H2: Control/evaluation divergence hypothesis

Among positions with similar engine evaluations, practical controllability metrics should vary materially, and those metrics should explain some outcome variance not captured by the scalar evaluation alone.

12.3 H3: Neutralisation hypothesis

Moves that materially reduce opponent decision-space entropy or near-optimal candidate count, without significant worsening of one’s own evaluation, should be associated with lower subsequent opponent counterplay.

12.4 H4: Critical-resource hypothesis

Interventions that remove a strategically critical resource should compress the opponent’s feasible plan-space more strongly than generic attacks on non-critical active pieces.

12.5 H5: Neutralisation–conversion separation

A player’s ability to resist a stronger engine should be statistically separable from their ability to convert advantageous positions into wins.

12.6 H6: Opponent-model hypothesis

The most effective neutralisation features should differ between human opponents, human-like engines such as Maia, and superhuman engines such as Stockfish.

13 Experimental Programme

The framework can be tested without requiring a new chess engine.

13.1 Dataset

A sufficiently large corpus should contain human games across multiple time controls and playing strengths, supplemented with controlled engine-vs-human or human-vs-Maia games where possible. Public Lichess game archives provide a natural starting point; Maia itself was developed using large collections of human games [5].

13.2 Engine analysis

For each position, run a fixed engine version with multi-PV analysis. Record at minimum:

- best evaluation and top- K evaluations;
- candidate-count within one or more evaluation thresholds;
- softmax decision-space entropy;
- evaluation volatility across top candidates;
- material and structural features;
- piece mobility and functional-mobility proxies;
- pawn-break availability;
- tactical motifs and forcing moves;
- identified critical-resource tags.

The 2026 decision-space entropy study provides a concrete template for using Stockfish multi-PV output to estimate decision-space complexity, although its own authors report limitations including lack of pre-registration and possible engine-assisted game contamination [9].

13.3 Move-level neutralisation score

For a move m , define a provisional neutralisation score:

$$N(m; S) = \alpha \Delta H_{opp} + \beta \Delta K_{opp} + \gamma \Delta R_{crit} - \delta \Delta \text{EvalLoss} - \eta \Delta \text{OwnRisk}, \quad (13)$$

where:

- ΔH_{opp} is the reduction in opponent decision-space entropy;
- ΔK_{opp} is the reduction in the number of near-optimal candidates;

- ΔR_{crit} is the reduction in critical resources;
- $\Delta EvalLoss$ measures the cost paid in objective engine evaluation;
- $\Delta OwnRisk$ measures increased vulnerability of the player’s own position.

The weights are unknown and must be estimated rather than asserted.

13.4 Ablation design

The strongest test would compare several move-selection policies:

1. engine-best move;
2. evaluation-preserving random choice among top candidates;
3. prophylactic heuristic only;
4. decision-space compression heuristic;
5. critical-resource heuristic;
6. combined neutralisation policy.

Outcomes should include engine score, game result, counterplay frequency, entropy trajectory, and conversion rate.

13.5 Human versus engine protocol

The cleanest human study would not simply ask whether people can beat Stockfish. It would measure whether training in the neutralisation framework changes the structure of their positions.

A possible intervention study could compare:

control group receiving conventional engine feedback *versus* training group receiving feedback on opponent intent, critical resources, decision-space compression, and control metrics.

The dependent variables would be post-training resilience, ACPL, blunder rate, entropy trajectory, and conversion performance.

14 What Is and Is Not Novel

A serious research claim must distinguish terminology from underlying ideas.

Table 3: Novelty assessment of the framework.

Element	Literature status	Position of this paper
Prophylaxis	Established positional principle	Historical foundation, not claimed as new
Restriction/restraint	Established	Integrated into the framework
Worst-piece improvement	Established	Used as a stabilisation mechanism
Squeezing small advantages	Established	Reframed through control versus evaluation
Human-like chess AI	Established through Maia and related work	Used to define opponent-model dependence

Decision-space entropy in chess	Emerging; recent 2026 empirical work exists	Related methodology; not claimed as invented here
Chess Engine Neutralisation	No established standard framework identified in the reviewed sources	Proposed integrated framework
Decision-space compression as prophylactic objective	Conceptual synthesis	Central research proposal; requires validation
Critical-resource targeting	Derived from classical strategic thinking, but formulated as a general heuristic here	Proposed operational vocabulary
Neutralisation–conversion separation	Practical distinction with testable formulation	Proposed research hypothesis
Control advantage separate from evaluation	Conceptual distinction	Proposed research dimension

The paper therefore does *not* claim that the underlying strategic ideas were invented from scratch. Its stronger and more defensible claim is that these ideas can be unified into a computational-opponent framework whose variables are measurable and experimentally testable.

15 Relation to the 2026 Decision-Space Literature

The timing of recent work on chess decision-space entropy is particularly relevant. Truong’s 2026 manuscript analyses 9,114 online games, using Stockfish multi-PV analysis to estimate a Luce/McFadden-style decision-space entropy and reporting a substantial association between entropy and human error [9]. The accompanying reproducibility repository documents both the methodology and important limitations, including engine-version sensitivity and possible engine-assisted games [10].

That work and this paper address different directions of the same broad construct.

Table 4: Two complementary decision-space questions.

Existing empirical direction	Neutralisation research direction
How does decision-space complexity relate to human error?	Can a player deliberately alter the opponent’s decision space?
What happens when decision entropy is high?	What happens when prophylaxis lowers decision entropy?
Measure decision space as an explanatory variable	Treat decision-space change as a strategic intervention

This convergence turns an informal chess philosophy into a potentially testable research programme. It also means novelty should be phrased carefully: the general notion of decision-space analysis in chess is now clearly part of the contemporary research conversation. The proposed novelty lies in the *direction of intervention* and the integration with prophylactic chess.

16 A Practical Human Thinking Routine

The framework can be reduced to a compact over-the-board routine.

1. **Intent:** If I pass, what does the opponent want?

2. **Critical resource:** Which piece, square, break, or exchange makes that plan work?
3. **Restriction:** Can I take that resource away?
4. **Safety:** Does my restriction create a tactical or structural concession?
5. **Improvement:** What is my worst piece after the opponent’s plan is neutralised?
6. **Probe:** What useful counterplay remains for the opponent?
7. **Conversion:** Can I now simplify, squeeze, or strike without reopening their resources?

This routine is deliberately compatible with positional play. It does not require a player to stop attacking; it changes the order in which attack and prevention are considered.

17 Limitations

The central limitation is empirical. The framework is currently stronger as a conceptual model than as a demonstrated predictive theory.

First, “effective decision space” is not uniquely defined. Different thresholds, depths, temperatures, engine versions, or candidate-selection rules can produce different metrics.

Second, engine evaluation is itself not a perfect ground truth. Stockfish can change strength and evaluation with version, hardware, time, and NNUE network. The official Stockfish project documents continued gains between releases [7].

Third, a reduction in candidate count is not automatically good. A move may accidentally reduce the opponent’s options because it blunders. Decision-space compression must therefore be conditioned on acceptable objective evaluation and acceptable own-position risk.

Fourth, human play cannot be reduced to engine choice distributions. Maia demonstrates that human move prediction is a distinct modelling problem [5]; decision-space metrics derived from engines should therefore be treated as structured proxies rather than literal descriptions of human cognition.

Fifth, the present practical observations are generated from repeated chess analysis rather than a pre-registered experiment. They are useful for hypothesis generation, not proof.

18 Discussion

The framework points toward a broader view of chess as **decision-space engineering**. Traditional chess language describes activity through pieces, squares, pawn structures, plans, threats, and exchanges. Engine-aware chess can add a further layer: the topology of the opponent’s choice set.

Under this view, a move can be strong for at least four different reasons:

1. it increases our static evaluation;
2. it improves our own future options;
3. it removes a critical opponent resource;
4. it reduces the diversity or quality of the opponent’s remaining choices.

The third and fourth reasons are the centre of this paper.

The framework also clarifies why a quiet move can be more strategically important than a forcing-looking move. A forcing-looking move may create activity but leave the opponent’s

decision space broad. A quiet prophylactic move may appear to do little while deleting an entire class of future plans.

This suggests a useful chess-design principle:

The strategic value of a move may lie partly in the alternatives it removes, not only in the advantages it visibly creates.

That proposition is philosophically close to classical prophylaxis, but its computational interpretation offers a path to measurement.

19 Conclusion

Chess Engine Neutralisation proposes a practical answer to a modern problem: how should a human think against computational opponents whose calculation is far beyond human reach?

The answer proposed here is not to pretend that the human can calculate like the machine. It is to recognise that chess strength is partly a function of the quality and structure of available choices. By applying prophylaxis deliberately, a player may be able to remove critical resources, reduce counterplay, compress the opponent’s effective decision space, stabilise the position, and improve incrementally.

The resulting framework is:

$$\boxed{\text{Neutralise} \rightarrow \text{Stabilise} \rightarrow \text{Probe} \rightarrow \text{Induce} \rightarrow \text{Strike} \rightarrow \text{Convert}} \quad (14)$$

Its most important conceptual distinctions are equally compact:

$$\boxed{\text{Evaluation advantage} \neq \text{Control advantage}} \quad (15)$$

and

$$\boxed{\text{Neutralisation skill} \neq \text{Conversion skill.}} \quad (16)$$

The first distinguishes how good a position is from how constrained it is. The second distinguishes the ability to prevent loss from the ability to realise advantage.

The paper’s most ambitious hypothesis is that prophylaxis can be reframed as an **intervention on decision space**. That hypothesis is now testable with modern multi-PV engine analysis, entropy estimation, human game corpora, and human-like models such as Maia. Recent 2026 work showing that decision-space entropy can be estimated at scale makes this research direction particularly timely [9].

The appropriate scientific posture is therefore neither “we invented prophylaxis” nor “the theory is proven.” The defensible position is stronger: classical positional ideas can be assembled into a coherent computational-opponent framework, the framework yields measurable constructs, and the resulting hypotheses are sufficiently concrete to be experimentally falsified.

If validated, this would offer chess theory a useful bridge between two traditions that are often discussed separately: **positional restraint** and **computational decision-making**. It would also provide a vocabulary for a phenomenon experienced by many human players but rarely formalised: sometimes the best way to survive a superhuman calculator is not to calculate harder, but to make the position give it less to work with.

A Glossary

Critical resource

The piece, square, pawn break, exchange, or coordination link whose preservation is disproportionately important to an opponent's plan.

Decision-space compression

A move-induced reduction in the opponent's strategically useful candidate set or decision entropy, conditional on acceptable evaluation and own-position safety.

Effective decision space

A subset or probability distribution over legal moves that captures moves with meaningful strategic value rather than merely legal moves.

Engine neutralisation

A strategy of constraining a computational opponent's useful resources so that its superior calculation produces fewer exploitable continuations.

Functional mobility

The set of moves or piece placements that preserve strategic purpose, as opposed to raw legal mobility.

Control advantage

A proposed second dimension of position quality describing stability, resilience, and restriction of opposing counterplay.

Neutralisation–conversion separation

The distinction between preventing the opponent from making progress and technically converting one's own advantage.

B Appendix: Research Pseudocode

INPUT: position S , side-to-move opponent = 0

1. Generate top- K candidate moves for 0.
2. Compute evaluation distribution $E = \{E_1, \dots, E_K\}$.
3. Estimate candidate count K_{active} within threshold τ .
4. Estimate entropy $H(E)$.
5. Tag strategic resources $R = \{\text{pieces, squares, pawn breaks, files, motifs}\}$.
6. For each candidate move m for the human player:
 - a. Apply $m \rightarrow S'$
 - b. Recompute 0's candidate set and entropy.
 - c. Estimate which resources in R disappeared or weakened.
 - d. Measure evaluation cost and own-risk change.
 - e. Score the move using a neutralisation objective.
7. Choose among moves with acceptable evaluation and risk.
8. After 0 replies, repeat the measurement.
9. Separate the game into:
 - neutralisation phase
 - stabilisation phase
 - probing/induction phase
 - conversion phase

OUTPUT: move, compression metrics, resource changes, conversion outcome

C Appendix: Reproducibility Recommendations

A future empirical version should publish:

- exact engine version and NNUE network identifier;
- hardware class and analysis time/depth;
- multi-PV setting and candidate threshold;
- entropy temperature and normalisation rules;
- game corpus and filtering criteria;
- engine-assistance/cheating filters;
- pre-registered hypotheses where feasible;
- code for feature extraction and statistical analysis;
- sensitivity results under alternative engine versions and thresholds.

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